

Peer Review Report

Manuscript under review

DeepCardio-RAG: A Multi-Modal Retrieval-Augmented Generation Framework for AI-Assisted Cardiac Diagnosis

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RECOMMENDATION: REJECT

resubmission encouraged, substantially rescoped

This was a harder call than the verdict suggests, and the reasoning matters more than the label.

The manuscript's own Table 14 is candid: four of the six deep-learning modules are marked *Not independently trained*, report-generation quality is marked *Not measured*, and Section 3.7 states plainly that the ventricular-arrhythmia module *uses random weights*. Of the two modules genuinely evaluated, both fall below the benchmarks they are placed beside: the flagship PTB-XL model reaches macro-AUC 0.772 against a published 0.925, and the arthritis ensemble reaches 0.811 against a published 0.88–0.93. That level of disclosure is unusual and creditable, and Section 9 of this report says so at length.

The problem is that the abstract, results prose, figure captions and conclusion contradict those tables. Section 4.3 states the echocardiography module *achieved MAE 4.1% on the held-out EchoNet test set*; Section 3.5 states the module never downloaded EchoNet and ran on synthetic input, and Table 14 marks it untrained. Section 4.5 states *the VFDB module in this prototype matches the benchmark* one sentence after stating it does not. Figure 5's caption asserts that ECG, VFDB and HD *exceed benchmarks*, when Table 14 shows one below benchmark and two untrained. These are not framing problems. They are results claimed that were never obtained.

Compounding this, the flagship demonstration triggers a Code Blue and defibrillation instruction from a randomly initialised network fed a synthetic signal, on a patient the system's own ECG module simultaneously reports as normal sinus rhythm at 72 bpm; the echocardiography module reports an ejection fraction of -0.1% , a physically impossible value, and that value propagates unchecked into the unified risk score. All three principal datasets are also misreported in ways this reviewer verified against the primary sources.

Once every unsupported claim is removed, what remains is a working software integration with four untrained modules plus one properly validated tabular classifier. That is a real engineering artifact but not a research contribution at this venue. Section 12 sets out two resubmission paths that this reviewer believes are genuinely viable.

1. Summary of the Submission

The manuscript presents DeepCardio-RAG, a multi-modal platform integrating six deep-learning modules covering ECG arrhythmia classification, LVEF estimation from echocardiography, heart-sound murmur detection, ventricular arrhythmia detection, heart-disease risk and arthritis risk. Module outputs feed a Cardio-Fusion scorer producing a 100-point risk index, and a Milvus-backed retrieval pipeline conditions a GPT-2 decoder that generates paired clinician and patient PDF reports. The system is delivered as a working prototype with a React dashboard, FastAPI backend and LangGraph orchestration, and is evaluated through a deployment session dated 15 June 2026.

What the manuscript does well

This section is longer than usual because the manuscript does several things most submissions in this area do not.

- **Table 14 is honest.** Marking four modules *Not independently trained* and the BLEU-4 result *Not measured*,

in the paper’s own headline results table, is the opposite of what most authors do. Section 5.1 goes further and states that BLEU-4 and hallucination-reduction figures *cited in earlier versions of this paper were not independently measured and have been removed*. Retracting one’s own prior numbers in print is rare and should be recognised.

- **The arthritis module is methodologically sound.** Section 3.9 identifies MCQ195 as a skip-pattern target-leakage source and excludes it. Table 9 reports SMOTE applied inside each cross-validation fold with the held-out split separated before resampling, repeated 5×3 stratified CV, and 95% confidence intervals, and states explicitly that these leakage-free numbers replace an earlier train-set evaluation. This is better practice than the majority of papers this reviewer sees in clinical tabular machine learning.
- **Section 4.10 discloses run-to-run variance** (macro AUC 0.80–0.86 across re-runs) and attributes it to best-checkpoint selection under minority-class scarcity, naming a class support as low as 19 beats. Most authors report the best run and say nothing.
- **The systems engineering is real.** Figures 2, 3 and 4 (inference flowchart, sequence diagram, server finite-state machine) are appropriate artifacts, the Milvus fallback path is explicitly designed, and the appendix pseudocode is concrete enough to inspect.

Core problem

The paper is two documents interleaved. One is a careful, self-critical prototype report. The other is a state-of-the-art claim built on results the first document says do not exist. A reviewer cannot accept the second on the strength of the first, and the second cannot simply be edited away, because removing it leaves too little behind.

2. What Was Actually Measured

Table 1 reorganises the manuscript’s own Table 14 to make the evidential position explicit.

Table 1: Evidential status of each claimed component, compiled from the manuscript’s Table 14, Section 3.5, Section 3.7 and Section 5.1.

Module	Trained?	Measured result	Benchmark	Assessment
Flagship ECG (Transformer-MoE), PTB-XL	Yes	AUC 0.772, F_1 0.467	0.925 (ref. 33)	Far below benchmark
ECG 1D-CNN, MIT-BIH beats	Yes	AUC 0.918, F_1 0.484	0.90 (ref. 3)	AUC comparable; F_1 0.484 is weak
CardioFusion (ECG path)	Yes	AUC 0.860, F_1 0.554	none given	No baseline to compare against
Arthritis NHANES ensemble,	Yes	CV AUC 0.792, test 0.811	0.88–0.93 (ref. 24)	Below benchmark, different task
Echo LVEF, EchoNet	No	none	MAE 4.1% (ref. 5)	§4.3 nonetheless claims the benchmark was achieved
PCG murmur, CirCor	No	none	78.5% (ref. 11)	Synthetic-signal confidence only
VFDB VF/VT	No (random weights)	none	98.7% (ref. 12)	§4.5 nonetheless claims it matches
Heart disease, UCI	No	none	AUC 0.85 (ref. 19)	Yet Table 8 reports 99.9% confidence
RAG report quality	—	Not measured	BLEU-4 >0.80	Correctly retracted in §5.1
RAG retrieval	—	recall@5 1.00, nDCG@5 0.99	0.958	Corpus size never stated; see M9

The central finding

Four of six modules were never trained. Report generation was never evaluated. Every module that *was* measured underperforms the benchmark it is placed beside. There is therefore no result in this manuscript that supports the abstract's claim of a platform "integrating six deep-learning modules," the conclusion's claim of "advancing the state of the art in three key dimensions," or Figure 7's award of a 6/6 capability score to this system against ≤ 2 for all prior work.

A capability score that counts the presence of a module, irrespective of whether that module has weights, is not an evaluation.

3. Prose Contradicting the Tables

Table 2: Statements in the running text and captions that contradict the manuscript's own tables.

#	Location	Claim made	Contradicted by
1	§4.3	"achieved MAE 4.1% on the held-out EchoNet test set, matching the benchmark of Ouyang et al."	Table 14 ("Not indep. trained"); §3.5 ("ran in synthetic demo mode, no EchoNet download")
2	§4.5	"The VFDB module in this prototype matches the benchmark."	The immediately preceding sentence; Table 7 ("benchmark not independently replicated"); §3.7 ("uses random weights")
3	Fig. 5 caption	"ECG, VFDB, and HD exceed benchmarks."	Table 14: ECG 0.772 vs 0.925; VFDB and HD untrained
4	Table 13	Arthritis "High Risk, 99.1% acc"	Table 9: held-out accuracy 0.765, AUC 0.811
5	Table 8	Heart disease "Confidence 99.9%, Disease Probability 99.9%"	Table 14: module not independently trained
6	§4.2	"Feature extraction accuracy 94.7% against the >90% benchmark of Hannun et al."	Table 14 gives PTB-XL AUC 0.772 / F_1 0.467; "feature extraction accuracy" is never defined
7	§3.10	CardioFusion is a "Cross-Modal Transformer + MMoE, 4 experts $\times$ 6 tasks, 15.8M parameters"	Eq. (13), Algorithm 4 and the appendix code all implement a fixed linear weighted sum with hard-coded λ
8	§3.13, Table 12	"384-dimensional BioBERT embeddings"	BioBERT-base produces 768 dimensions; §3.11 sets <code>EMBEDDING_DIM</code> 768
9	§1, contributions	"the first open-source research prototype"	Data Availability: "source code ... available from the corresponding author on reasonable request"
10	Table 14	Inference on "Colab GPU (T4)"	§4 header and Table 12: Colab CPU , 57.39 s
11	§3.6, Table 13	Combined inference 0.09 s	Table 12: system inference 57.39 s
12	Fig. 6 caption	"Green >80%, yellow 60–80%, red <60%"	The rendered heatmap uses a blue-to-red colormap containing no green
13	Fig. 5 caption	"0 = not measured"	The vertical axis begins at 40, so a zero cannot be displayed
14	Table 3	$P = 100$ patches, $C = 12$ classes	§3.4 and Algorithm 1: $P = 50$; the AAMI task has 5 classes
15	Table 12	Arthritis predictor is "GradientBoosting, 100 est., depth 4"	§3.9 and Table 9: a five-model stack with a logistic-regression meta-learner

Items 1 and 2 require separate comment

The distinction between overclaiming and misreporting matters here. Writing "our architecture follows the design of Ouyang et al., who report MAE 4.1%" would be acceptable. Writing "achieved MAE 4.1% on the held-out EchoNet test set" asserts that the authors ran an evaluation and obtained that number. Section 3.5 establishes that no EchoNet data was ever downloaded. The same applies to Section 4.5,

where the claim that the module “matches the benchmark” appears in the sentence immediately after the disclaimer that it does not.

This reviewer’s reading is that these are legacy sentences from an earlier draft that survived the honest revision of the tables, rather than deliberate misstatement. Either way they must be removed before this manuscript is sent anywhere, because a reader without the tables in front of them will take them at face value.

4. Clinical Safety Analysis of the Demonstration Case

The manuscript’s flagship result is patient ECG-20260615-0836, presented in Tables 5 and 13, in Section 4.4 and in the conclusion. Table 3 traces what the system did.

Table 3: Provenance of each finding for demonstration patient ECG-20260615-0836.

Module	Reported finding	Input	Model status and assessment
ECG 1D-CNN	Normal Sinus Rhythm, conf. 19.1%	PTB-XL record	Trained. 19.1% on a 5-class problem is below the 20% chance level
ECG image CNN	Unclassifiable (Q), conf. 28.8%	Demo image	Q has no entry in the paper’s own class counts (see M6)
Echo 3D-CNN	HFrEF, EF = −0.1%	Synthetic tensor	Untrained. Negative ejection fraction is physically impossible and violates the paper’s own $\hat{y}_{EF} \in [0, 100]\%$
PCG Mel-CNN	Murmur Present, 44.4%	Synthetic signal	Untrained
VFDB 1D-CNN	VF/VFL Critical , 32.5%	Synthetic signal	Random weights
Heart disease	HIGH Risk, 99.9%	Demo features	Untrained. 99.9% from an untrained network is a random number
Arthritis	High Risk, “99.1% acc”	Demo features	Trained, but Table 9 gives 0.765 accuracy
System output: CardioFusion 70/100 HIGH RISK; CRITICAL combined alert; guidelines issued: “Ventricular Fibrillation/Flutter detected — Activate code blue” and “Immediate defibrillation, CPR if pulseless, Epinephrine every 3–5 min (AHA ACLS)”			

M1. The demonstration case is a patient-safety failure presented as a success

Four observations, in ascending order of seriousness.

(a) The alert is internally impossible. Table 5 reports the same patient at 72 bpm in normal sinus rhythm with PR 160 ms, QRS 90 ms and QTc 410 ms, all normal. Ventricular fibrillation is incompatible with an organised sinus rhythm at 72 bpm and with a conscious patient. The system therefore issued a resuscitation directive that its own concurrent ECG analysis contradicts, and the manuscript does not notice.

(b) The alert originates from noise. The VF/VFL detection came from a randomly initialised network processing a synthetic signal. The paper describes this in Sections 3.6, 3.7 and 4.4 and still presents the resulting CRITICAL alert as a demonstration of correct operation, with a 0.09 s inference time quoted as though latency were the relevant property.

(c) The output is directive, not advisory. “Activate code blue,” “Immediate defibrillation,” “Epinephrine every 3–5 min,” “administer reperfusion therapy,” “coronary angiography.” These are treatment orders, not flagged observations. The abstract’s disclaimer that the system “has not been validated for clinical deployment” does not neutralise a system whose demonstrated output instructs a clinician to defibrillate. Coronary angiography, recommended in Table 8 on the basis of an untrained model’s 99.9% output, carries genuine procedural risk.

(d) A physically impossible value propagates silently. An ejection fraction of -0.1% should be caught by a validity check before it reaches a risk score. It is instead used to set a HFrEF flag, and the conclusion cites “reduced LVEF (HFrEF) and detected murmur” as evidence that CardioFusion “demonstrated risk stratification aligned with clinical intuition.” No range check exists anywhere in the pipeline.

The revision must add input-validity and physiological-plausibility gating, must suppress the outputs of untrained modules rather than reporting them with confidences, must reframe every recommendation as an observation requiring clinician confirmation, and must add explicit discussion of automation bias and of the system’s regulatory status as software as a medical device.

M2. The retrieval demonstrably fails at the task RAG is meant to solve

For this patient, whose only abnormality is an ST wave change, the pipeline retrieved: STEMI reperfusion guidance, atrial fibrillation anticoagulation, PVC-induced cardiomyopathy, ST depression, and QTc prolongation. Three of the five are irrelevant: the patient has no ST elevation, is in sinus rhythm rather than AF, and has a normal QTc of 410 ms.

Retrieving STEMI reperfusion guidance for a patient without ST elevation is precisely the failure mode retrieval augmentation is introduced to prevent. The manuscript presents this as successful grounding.

M3. The vector store does not contain what the paper says it contains

Section 3.13 and Table 12 both state that the Milvus collection is `arthritis_patients` with 5,758 encoded records. The NHANES arthritis table has 5,756 records (Table 2). The collection being queried is therefore the arthritis patient table, not a clinical guidelines corpus.

Five AHA/ACC cardiac guidelines cannot be retrieved by cosine similarity from a collection of NHANES survey records. Figure 2 shows the likely resolution: the inference flowchart contains an explicit branch, “Milvus available? no: demo guidelines.” The five guidelines reported in Section 4.2 are plausibly the hardcoded fallback list rather than retrieved content. If so, the RAG pipeline was not exercised at all in the demonstration, and the paper’s central architectural claim is undemonstrated.

The authors must state the guideline corpus size, its collection name, its provenance, and whether the reported retrieval executed against Milvus or fell through to the fallback branch.

5. Dataset Integrity Findings

This reviewer checked the three principal datasets against their primary sources. All three are misreported.

M4. The PTB-XL class labels are attached to the wrong counts

Section 3.12 and Section 4.2 state: “Real ECG Records 21,430, NORMS 9,528, CD 5,486, HYP 5,250, MI 7,562.”

The published PTB-XL superclass distribution [R1] is NORM 9,528, MI 5,486, STTC 5,250, CD 4,907, HYP 2,655, over 21,837 records (21,799 in v1.0.3).

Superclass	Manuscript	PTB-XL, actual
NORM	9,528	9,528 ok
MI	7,562	5,486
STTC	<i>absent</i>	5,250
CD	5,486	4,907
HYP	5,250	2,655
Total records	21,430	21,837 (21,799 in v1.0.3)

The manuscript’s CD count is PTB-XL’s MI count; its HYP count is PTB-XL’s STTC count; STTC is omitted entirely; and 7,562 corresponds to no PTB-XL class. The four listed counts sum to 27,826,

exceeding the stated 21,430 records, and the paper presents them as a partition rather than as multi-label statistics.

Since these figures are described as confirmed by the dashboard and are used to characterise the corpus against which the flagship AUC of 0.772 was obtained, the authors must establish which PTB-XL version was used, how records were selected to reach 21,430, and whether the label mapping in the code matches the counts printed in the paper. If the labels are mis-mapped in the code rather than only in the text, the flagship result is invalid and the unusually low AUC would have an explanation.

M5. The UCI Cleveland file used contains duplicated records

Section 3.8 and Table 2 report “1,025 patients” for the UCI Cleveland Heart Disease dataset. The original Cleveland cohort is 303 patients [R2]. The 1,025-row file circulating on Kaggle is the 303-row original expanded with more than 700 duplicate entries; this is documented in the literature together with the observation that validating on it produces inflated and unreliable metrics, because identical rows appear on both sides of any random split [R3].

The reported prevalence of 51.3% is also inconsistent with the original Cleveland prevalence of roughly 46%, which is what duplication of a non-uniform subset would produce.

Because this module was never trained (Table 14), no current result depends on the file. But if the module is trained in a future version, deduplication to the 303 unique records, or a patient-level split, is mandatory.

M6. The MIT-BIH image class counts merge two classes

Table 4 reports “N 90,589, S 2,779, V 7,236, F 8,841 (5 classes),” totalling 109,445.

The standard MIT-BIH AAMI beat distribution is N 90,589, S 2,779, V 7,236, F 803, Q 8,039, totalling 109,446. Note that $803 + 8,039 = 8,842$. The manuscript’s “F 8,841” is the F and Q classes summed, off by one, with Q then omitted from the listing while the caption still says five classes. The stated total of 109,445 is likewise one short of 109,446.

This is not merely typographic. Section 4.2 reports the image classifier predicting “Unclassifiable (Q)” with 28.8% confidence, a class absent from the paper’s own class table. And Section 4.1 states that “the 20,000-beat demo sample used for live training was 100% Normal (N) class,” meaning a five-class classifier was trained on a single class. That sentence alone would explain the weak F_1 of 0.484 and needs to be confronted rather than reported in passing.

M7. Further dataset accounting problems

- **Arthritis record count** appears as 5,756 (Table 2, §3.9, Table 4), 3,000 (Table 9, “stratified subsample”), and 5,758 (Table 12, encoded records). Subsampling to 3,000 discards nearly half the available data with no stated justification. Reference 32, “Heart Disease Dataset—3000 Records, Kaggle (2025),” is never cited in the text and its relationship to the subsample is unexplained.
- **Feature count** appears as 8 predictors plus interactions (§3.9), 17 engineered features (Table 9), 12 features (Table 4), and 12 rows in Table 10.
- **VFDB rhythm distribution** in §4.1 lists counts summing to roughly 148 while the stated total is 142 dangerous events, and includes N (≈ 62), which is not a dangerous event.
- **CirCor benchmark** is given as outcome metric 0.789 in §2.3 and as 78.5% in Table 14.

6. Further Major Concerns

M8. CardioFusion as described and CardioFusion as implemented are different systems

Section 3.10 describes a “Cross-Modal Transformer + Multi-gate Mixture-of-Experts (MMoE) architecture (4 experts $\times$ 6 tasks, shared 512-dim space, 4-layer Transformer fusion, 15.8M total parameters, model size 60.4 MB).”

Equation (13), Algorithm 4 and the appendix code all implement

$$R = 100 \sum_{m=1}^6 \lambda_m p_m^{\text{risk}}, \quad \lambda = (0.25, 0.25, 0.20, 0.15, 0.10, 0.05),$$

a fixed linear combination with hardcoded weights and no learned parameters at all. A 15.8M-parameter learned transformer and a six-term weighted sum are not the same object. The 15.8M-parameter model also does not appear in Table 14, so it was presumably never trained either.

Three further problems follow. The weights are asserted with no derivation, no sensitivity analysis and no clinical justification for ranking echocardiography above ventricular arrhythmia. Table 13 lists *seven* module outputs against six weights, and the manuscript does not say which is dropped. And the reported score of 70/100 is not reproducible from Equation (13) using Table 13’s confidences under any obvious mapping; this reviewer obtained 58.5 assuming the HFrEF flag maps to $p = 1.0$ and the ECG confidence maps directly. Algorithm 4 line 3, “Map $\hat{y}_m \rightarrow p_m^{\text{risk}}$,” is the missing definition and must be specified.

M9. The retrieval metrics are uninformative as reported

Table 14 reports recall@5 of 1.00 and nDCG@5 of 0.99 for RAG retrieval. If the guideline corpus contains five documents, as Section 4.2 implies, then retrieving the top five yields recall@5 = 1.00 by construction, for any embedding whatsoever. The corpus size is never stated.

Meaningful evaluation requires the number of guideline chunks, the number of queries, how relevance judgements were produced and by whom, and comparison against a lexical baseline on the same query set. The stated comparator, “0.958 (lexical TF-IDF),” has no accompanying description.

M10. The arthritis module solves a different problem from the one motivating it

This is the strongest module in the paper and deserves precise criticism.

The prediction target is MCQ160A, self-reported doctor-diagnosed arthritis in a household survey. That population is overwhelmingly osteoarthritis. The manuscript’s entire motivation, in Section 1, Section 2.6 and the title, concerns *rheumatoid* arthritis and its two-fold cardiovascular risk, citing the ACR/EULAR 2010 criteria and Tian et al.’s RA-from-biomarkers work at AUC 0.88–0.93.

Excluding MCQ195 was the correct decision, but its consequence is that the module can no longer identify RA at all. Comparing 0.811 on self-reported any-arthritis against 0.88–0.93 on serologically defined RA compares different tasks, and the cardiovascular comorbidity argument does not transfer to osteoarthritis. Either the framing changes to arthritis burden generally, with the RA-specific cardiovascular motivation removed, or a cohort with genuine RA labels is required.

Secondary points: three collinear age encodings (age_risk 15.7%, age_squared 12.8%, Age 11.5%) account for 40% of total importance, indicating a model that is largely an age model with engineered redundancy inflating apparent feature richness; the held-out AUC of 0.811 lies outside the CV confidence interval of [0.781, 0.803] and this warrants comment; and InflammationProxy is never defined, is described as an “inflammatory marker” with mean 0.035, yet takes only the values 0 and 0.5 in Table 11.

M11. Table 11 contradicts the model it illustrates

Given that age carries 40% of feature importance, the sample predictions are hard to reconcile: PT-0003 (male, 72, BMI 28.9, SBP 140) is labelled Normal while PT-0006 (female, 61, BMI 35.7, SBP 118) is

labelled High. PT-0007 has missing BMI and SBP yet still receives a prediction, with no indication of how imputation was handled at inference. None of this is discussed.

M12. The theoretical apparatus adds nothing and contains an error

- **Theorem 1** is the Cauchy–Schwarz inequality applied to a dot product. It is not a theorem, it is not specific to this architecture, and the stated consequence does not follow: the bound $M^2/\sqrt{d_k}$ grows without limit as M grows, so it prevents nothing.
- **Theorem 2** states the universal approximation property for MoE without proof. As written it restates the standard result for a single MLP, since one expert with a hidden layer already suffices and the condition $K \geq 2$ is irrelevant. It is not the authors’ result and carries no attribution.
- **Lemma 1** is incorrect. From $F(\mathbf{x}) = \mathbf{x} + H(\mathbf{x})$ one obtains $\partial\mathcal{L}/\partial\mathbf{x} = (\partial\mathcal{L}/\partial F)(1 + \partial H/\partial\mathbf{x})$, and the conclusion that the gradient magnitude is at least $|\partial\mathcal{L}/\partial F|$ requires $\partial H/\partial\mathbf{x} \geq 0$, which is not guaranteed. The loss symbol is also missing from the statement as printed.
- **Proposition 1** states that a positively weighted sum is monotone in its arguments.

In a paper where four modules are untrained, presenting universal approximation theorems as contributions is a mismatch of form and substance. This material should be removed.

M13. Architectural specification is internally inconsistent

- Section 3.4 states the ECG input is $\mathbf{X} \in \mathbb{R}^{12 \times P \times 50}$ with $P = 50$, from MIT-BIH beats of 187 samples on Lead II plus modified V1. MIT-BIH is a two-lead database, not twelve. And 187 samples cannot be partitioned into non-overlapping patches of 50.
- Algorithm 1 requires a 12-lead input and $C = 12$ classes; the AAMI task has 5. The appendix code confirms `nn.Linear(d, 12)`.
- Four different shared embedding dimensions appear: 256 (module embeddings, §3.4), 384 (retrieval space, §3.13), 512 (CardioFusion, §3.10) and 768 (EMBEDDING_DIM, §3.11).
- Table 12 describes the ECG encoder as “1D-CNN, 3 blocks, 256-dim” while §3.4 and Algorithm 1 specify four blocks.

M14. The deployment is a demo session, not an experiment

The phrase “the June 15, 2026 deployment” appears roughly fifteen times as though it were a controlled evaluation. It was a single session on Google Colab CPU with Milvus exposed through a public `bore.pub` tunnel. There is no repetition, no seed, no split index, no environment specification, and no separation between development and evaluation. Single-session demonstrations cannot support performance claims.

Two further points. Exposing a vector store containing patient-labelled records through a public tunnel warrants comment even for de-identified public data, and the appendix code hardcodes `host="localhost"`, contradicting the tunnel deployment. And Limitation 1’s assertion that A100 hardware “is expected to reduce latency below 2.0 s” proposes a 28-fold speedup with no profiling breakdown; 57.39 s is more plausibly dominated by model loading than by compute, in which case GPU migration will not deliver it.

7. Minor Concerns

Table 4: Minor issues, in manuscript order.

ID	Location	Issue
m1	Author contributions	Contributions are stated for three authors (N.R.R., D.N., S.S.) but four are listed. N. Soumya, the only author at a medical college, has no stated contribution, while clinical domain expertise and medical literature review are attributed to an author in a Department of AI and Data Science. This should be corrected, both for accuracy and in fairness to the clinical co-author.
m2	Affiliations	The first author carries two affiliations (SR University and KL Education Foundation) with an @klh.edu.in address but an SR University correspondence line, unexplained.
m3	Title	Lists “Arthritis Risk Classification” alongside cardiac modalities; given M10, the title should not imply RA.
m4	§1, contributions; §2.9	“the first open-source research prototype” and “No published system integrates all major non-invasive cardiac modalities.” Priority claims of this scope require a documented search strategy, not a table of six hand-picked comparators.
m5	Table 15, Fig. 7	The comparison set contains four systems that are single-modality by design and two general medical LLMs. Actual multi-modal cardiac systems and medical multimodal foundation models are absent, which makes the 6/6 versus ≤ 2 result an artifact of comparator selection.
m6	Ref. 12	Cited throughout for “13-layer 1D-CNN, 98.7% for VF/VT on MIT-BIH VFDB,” but the reference given is Acharya et al., <i>A deep convolutional neural network model to classify heartbeats</i> , which is the heartbeat paper, not the shockable-rhythm paper. Verify and correct.
m7	Ref. 19, §3.8	Cited as the source of the UCI Cleveland dataset. The dataset originates with Detrano et al. (1989); a 2020 JACC paper is not its provenance.
m8	Ref. 22	Author is Aviña-Zubieta, J.A., not “Avina-Zubieta, C.”
m9	Refs. 47–59	Thirteen references (ResNet, Attention, BERT, GPT-2, BioBERT, Goodfellow, Topol, MIMIC-III and others) that support no specific claim in the text. Reference padding.
m10	§4.1	“Hematology: mean BMI 26.7, SystolicBP ≈ 120 , DiastolicBP 68.” None of these are hematology measurements.
m11	§4.1 vs Table 8	“Disease-positive chest pain types: typical angina (strongest predictor)” conflicts with Table 8, where chest pain type 0 is labelled “Asymptomatic.” In the standard UCI encoding 0 is typical angina. The encoding is inconsistent between sections.
m12	Table 9 footnote	“These leakage-free numbers replace an earlier train-set evaluation.” The right disclosure, but it belongs in the main text, and the superseded numbers should be stated so readers of the earlier version can reconcile.
m13	§3.11	MILVUS_PORT 19530 while the same section gives the tunnel as port 38467 and §4’s header repeats 19530.

ID	Location	Issue
m14	Appendix code	<code>nn.Linear(d,12)</code> for a 5-class task; <code>host="localhost"</code> contradicts the tunnel deployment; no seeds; no training loop for four of six modules.
m15	Statements	No ethics or IRB statement. All data are public and de-identified, which likely exempts the work, but this should be stated. No statement on regulatory classification given the directive outputs.
m16	Fig. 5	Mixes AUC, F_1 , accuracy, confidence and a synthetic-signal probability on one “Performance (%)” axis. These are not commensurable quantities.
m17	§5.3, Limitation 3	“may eliminate residual hallucinations” overstates what a larger model achieves; hallucination is reduced, not eliminated, and the paper has not measured it.

8. Reporting Completeness

Criterion	Status	Comment
Datasets identified and public	Yes	All six named with sources
Dataset statistics accurate	No	Three of three checked are misreported (M4–M6)
All modules trained	No	Four of six untrained
Splits and seeds reported	Partial	Only for the arthritis module
Repeated runs with variance	Partial	Arthritis only; §4.10 discloses ECG variance without quantifying it
Confidence intervals	Partial	Arthritis only
Statistical testing vs baselines	No	None
Calibration analysis	No	Absent, though confidences drive alerts
Leakage controls	Partial	Exemplary for arthritis; duplicated UCI file elsewhere
Generation quality measured	No	Correctly disclosed as not measured
Retrieval corpus characterised	No	Size and provenance never stated
Code publicly available	No	“On reasonable request,” contradicting the open-source claim
Model weights available	No	Not mentioned
Hardware and runtime consistent	No	CPU vs GPU contradiction; 57.39 s vs 0.09 s
Input validity checking	No	$EF = -0.1\%$ propagates to the risk score
Ethics / regulatory statement	No	Absent despite directive clinical outputs

One criterion of sixteen is met in full; four partially, and all four are carried by the arthritis module alone.

9. Scoring

Criterion	Score (1–5)	Justification
Originality of the idea	3	Unified multi-modal cardiac RAG is a reasonable and timely concept
Technical execution	1	Four of six modules untrained; CardioFusion described but not implemented as described
Experimental rigour	1	Single demo session; no seeds, CIs or testing outside the arthritis module
Accuracy of reporting	1	Prose contradicts the paper’s own tables in fifteen places
Dataset handling	1	All three datasets checked are misreported; duplicated UCI file
Clinical safety	1	Code Blue issued from random weights on synthetic input; no validity gating
Comparison with prior work	2	Comparator set selected in a way that guarantees the result
Transparency and honesty	4	Table 14, §4.10 and the §5.1 retraction are genuinely creditable
Reproducibility	1	One of sixteen criteria met
Clarity of presentation	3	Well organised and readable; figures need work
Overall	1.5 / 5	Reject, resubmission encouraged after rescoping

10. Questions for the Authors

- Q1. Section 4.3 states the echo module achieved MAE 4.1% on the held-out EchoNet test set, while Section 3.5 states EchoNet was never downloaded. Which is correct?
- Q2. Section 4.5 states the VFDB module matches the 98.7% benchmark and, in the preceding sentence, that it does not. Which is correct?
- Q3. On what basis does Figure 5’s caption state that VFDB and HD exceed their benchmarks?
- Q4. Where does the arthritis figure of 99.1% in Table 13 come from, given Table 9’s held-out accuracy of 0.765?
- Q5. Was the CardioFusion 15.8M-parameter Cross-Modal Transformer implemented and trained, or is Equation (13) the actual implementation?
- Q6. How is the reported score of 70/100 obtained from Equation (13) and Table 13? Please give the mapping $\hat{y}_m \rightarrow p_m^{\text{risk}}$ and the six values used.
- Q7. How many documents are in the guideline corpus, and did the reported retrieval execute against Milvus or fall through to the demo-guideline branch of Figure 2?
- Q8. Which PTB-XL version and record selection yields 21,430 records, and does the code’s label mapping match the class counts printed in Section 3.12?
- Q9. Was the 1,025-row UCI file deduplicated to the 303 unique Cleveland records?
- Q10. Section 4.1 states the 20,000-beat live training sample was 100% Normal class. How was a five-class classifier trained on one class, and does the reported F_1 of 0.484 come from that run?
- Q11. What input-validity checks exist? Specifically, how did $EF = -0.1\%$ reach the risk scorer?
- Q12. What is the intended regulatory classification of a system whose demonstrated output includes “Immediate defibrillation” and “administer reperfusion therapy”?

11. Confidential Comments to the Editor

On the nature of the problem

This submission differs from the usual overclaiming case. The authors have documented their own limitations in Table 14, Section 4.10 and Section 5.1 more candidly than most, including retracting BLEU-4 and hallucination figures from an earlier version. That is to their credit and this reviewer wants it on the record.

The difficulty is that the abstract, results prose, figure captions and conclusion were evidently not reconciled with those tables. Section 4.3's claim of an achieved EchoNet MAE and Section 4.5's claim of a matched VFDB benchmark are statements of results that the same manuscript establishes were never obtained. The most probable explanation is that these are legacy sentences from an earlier draft that survived the honest revision of the tables, and the editor may wish to put that reading to the authors rather than treating it as deliberate misstatement. Either way they cannot remain.

On clinical safety

This reviewer would ask the editor to give particular weight to Section 4 of this report. A system that issues "Activate code blue" and "Immediate defibrillation" from a randomly initialised network on a synthetic signal, for a patient its own module concurrently reports as being in normal sinus rhythm at 72 bpm, should not be published as a working demonstration in any form. The negative ejection fraction propagating unchecked into the risk score compounds this. A journal-level position on the outputs of clinical decision-support prototypes may be warranted independently of the fate of this manuscript.

On the balance of the decision

Rejection rests on the position after correction, not on the number of errors. Removing every unsupported claim leaves: a software integration in which four of six modules have no trained weights; a PTB-XL model at AUC 0.772 against a 0.925 benchmark, trained on a corpus whose reported label distribution does not match PTB-XL; and one well-validated tabular classifier at AUC 0.811 on self-reported arthritis, a task that does not support the paper's rheumatoid-arthritis framing. That is not a contribution at this venue.

However, this reviewer does not regard the work as unpublishable, and Section 12 sets out two routes that would be. If the editor prefers a revision cycle to rejection, the minimum conditions would be: training all six modules or removing the untrained ones entirely from every table, figure and claim; correcting the three dataset misreports; implementing input-validity gating and reframing all directive outputs as advisory; and reconciling the prose with the tables line by line.

Reviewer confidence: High on machine-learning methodology, evaluation practice and the dataset findings, which were verified against PhysioNet and the published literature. Moderate on cardiology; a clinical co-reviewer should assess Section 4 of this report and the guideline content of the manuscript's Section 4.2 independently.

12. Two Viable Resubmission Paths

The engineering here is real and the honesty is real. Both are worth building on.

Path A: a systems and architecture paper

Reframe entirely as an engineering contribution and target a health-informatics systems or software venue. The claim becomes: *here is a reference architecture for orchestrating heterogeneous clinical models behind a retrieval-grounded reporting layer, with an open implementation*. Under this framing, untrained modules are acceptable as interface placeholders provided they are labelled as such everywhere and their outputs are suppressed rather than scored. What must then be evaluated is the architecture: per-stage latency profiling, failure and fallback behaviour, throughput under load, and the finite-state recovery properties already sketched in Figure 4. Publish the repository. Remove every accuracy claim, Table 15 and Figure 7. This is achievable within weeks and the artifact already largely exists.

Path B: a focused arthritis-risk paper

The NHANES module is the only properly validated component and the methodology behind it is good. A standalone paper would need: the full 5,756 records rather than a 3,000 subsample; the framing corrected from rheumatoid arthritis to self-reported arthritis burden, with the cardiovascular comorbidity argument adjusted accordingly; the collinear age encodings consolidated and importances recomputed; calibration analysis added, since the intended use is risk stratification; and comparison against a plain logistic regression and a plain gradient-boosted tree, because a five-model stack plus a BERT-MoE must justify its complexity against

those. If the stack does not beat logistic regression by a meaningful margin, that is itself a publishable and useful finding [R8].

If the multi-modal paper is to be pursued

Train the modules. Four public datasets with published baselines are available and the compute required is modest by current standards. Then evaluate what the paper actually claims is novel, which is neither the individual modules nor their sum, but the fusion and the grounding. Two clean experiments would carry it: does CardioFusion beat the best single modality on a held-out multi-modal cohort, and does retrieval grounding measurably reduce factual error against an unretrieved GPT-2 baseline on a rated report set? Either one on its own would be a stronger paper than the present submission.

References Consulted for This Review

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- [R2] R. Detrano et al., “International application of a new probability algorithm for the diagnosis of coronary artery disease,” *American Journal of Cardiology*, vol. 64, pp. 304–310, 1989. Origin of the 303-record Cleveland cohort.
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- [R4] N. Strodthoff, P. Wagner, T. Schaeffter and W. Samek, “Deep learning for ECG analysis: benchmarks and insights from PTB-XL,” *IEEE Journal of Biomedical and Health Informatics*, vol. 25, no. 5, pp. 1519–1528, 2021. The 0.925 macro-AUC benchmark cited by the manuscript as reference 33.
- [R5] D. Ouyang et al., “Video-based AI for beat-to-beat assessment of cardiac function,” *Nature*, vol. 580, pp. 252–256, 2020.
- [R6] J. Oliveira et al., “The CirCor DigiScope dataset: from murmur detection to murmur classification,” *IEEE Journal of Biomedical and Health Informatics*, vol. 26, no. 6, pp. 2524–2535, 2022.
- [R7] P. Lewis et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” *NeurIPS*, vol. 33, pp. 9459–9474, 2020.
- [R8] “Stronger baseline models: a key requirement for aligning machine learning research with clinical utility,” arXiv:2409.12116, 2024. Demonstrates that logistic regression matches or exceeds reported deep models on the Cleveland task; directly relevant to Path B.
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- [R10] Z. Ji et al., “Survey of hallucination in natural language generation,” *ACM Computing Surveys*, vol. 55, no. 12, 2023. Relevant to the retracted hallucination-reduction claim and to what a proper measurement would require.

End of review report